THEOPHYLLINE

Narrow therapeutic index

Many interactions

CHEMICAL NAME 1,3-dimethyl-3,7-dihydro-1H-purine-2,6-dione
CLASS Methylxanthine bronchodilator
TRADE NAMES Uniphyllin Continus · Nuelin SA · Theo-24 · Theodur

THERAPEUTIC USES / INDICATIONS

- Chronic asthma (add-on) and COPD (when inhaled therapy inadequate).
- Apnoea of prematurity (caffeine preferred).

ADMINISTRATION

- **PO** modified-release tablets swallowed whole, every 12 h.
- Brands not interchangeable — prescribe by brand.
- Monitor plasma levels 4–6 h after dose (MR) at steady state.

DOSAGES

- **Adult:** 200–300 mg MR every 12 h (adjust by levels).
- **Target level:** 10–20 mg/L (55–110 µmol/L) (5–15 mg/L often sufficient).
- Reduce in heart failure, liver disease, elderly, with interacting drugs.
- Smokers need higher doses.

PHARMACOKINETICS

- **Peak:** 4–7 h (MR).
- Metabolised in liver by CYP1A2 (induced by smoking; inhibited by ciprofloxacin, erythromycin).
- Excreted in urine. **t½:** 8 h adults (4–5 h smokers; longer in HF, liver disease).

PHARMACODYNAMICS

- Inhibits **phosphodiesterase** → ↑ cAMP → bronchodilation; antagonises adenosine receptors.
- ↑ diaphragm contractility, respiratory drive; mild diuresis.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, vomiting, abdominal discomfort.
- Headache, insomnia, irritability.
- Palpitations, tachycardia.

ADVERSE EFFECTS (SERIOUS)

- Toxicity (>20 mg/L): persistent vomiting, arrhythmias, seizures, hypokalaemia, hyperglycaemia.
- Can be fatal; seizures may occur without warning signs.

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity to xanthines; acute porphyria.
- **Caution:** cardiac arrhythmias, epilepsy, peptic ulcer, hyperthyroidism, liver disease, heart failure, elderly, fever.

KEY INTERACTIONS

- **↑ levels:** ciprofloxacin, clarithromycin, erythromycin, fluvoxamine, cimetidine, allopurinol, oral contraceptives, stopping smoking.
- **↓ levels:** smoking, rifampicin, carbamazepine, phenytoin, St John's wort.
- **β2 agonists, steroids, diuretics:** hypokalaemia.

NURSING MANAGEMENT

- Check levels when starting, changing dose or adding interacting drugs.
- Monitor HR, rhythm, K^+, GI symptoms.
- Ask about smoking status changes.
- Same brand each time.

PATIENT EDUCATION

- Swallow tablets whole; take every 12 h.
- Tell your doctor if you start or stop smoking.
- Report vomiting, palpitations, shaking.
- Limit caffeine (coffee, tea, cola).

KEY POINTS

- Methylxanthine; PDE inhibitor.
- CYP1A2 interactions (ciprofloxacin, smoking).
- Target 10–20 mg/L.
- Toxicity: vomiting, arrhythmias, seizures.